Supplementary material for

Feature selection and relevance of radiomic features for reperfusion prediction

Petra Nemcekova, Jiri Chmelik, Tomas Vicar, Mahsa Mojtahedi, Henk A. Marquering, Roman Jakubicek

Supplementary tables

Table 1: Patient selection in the two cohorts. MR CLEAN NO-IV was used for model development and cross-validation, the MR CLEAN Registry as the independent external cohort. Entries are numbers of patients; an em dash marks an exclusion step that does not apply to that cohort. eTICI, expanded Thrombolysis in Cerebral Infarction score; the outcome modeled throughout this article is near-complete or complete reperfusion, defined as a final eTICI score of 2C or 3. Slice spacing refers to the through-plane spacing of the non-contrast CT acquisition.

Step	MR CLEAN NO-IV	MR CLEAN Registry
Initial cohort	539	2597
Excluded		
Missing eTICI score	—	16
Missing thrombus segmentation	—	575
Slice spacing > 1 mm	—	100
Feature extraction failure	243	—
Manual segmentation quality review	155	*not performed*
Final analysis cohort	**141**	**1906**
Reperfusion eTICI 2C–3, n (%)	82 (58.2%)	—

Notes to Table 1. The manual review of segmentation quality was carried out on the MR CLEAN NO-IV cohort only. No segmentation was reviewed or rejected in the MR CLEAN Registry cohort, so the two cohorts differ in segmentation quality control as well as in population. This is a property of the design and is the single largest asymmetry between the development and the external cohort. The review was blinded to the eTICI score: it judged the quality and the anatomical placement of the segmentation only, so the 155 exclusions it produced cannot have been informed by the outcome. Of the 141 patients that remain, 82 (58.2%) reached eTICI 2C–3; the proportion varies across the five outer folds (0.448, 0.714, 0.536, 0.679, 0.536), so in the first fold the majority class is the absence rather than the presence of near-complete or complete reperfusion.

Table 2: Hyperparameter search spaces of the four pipelines. LASSO, ℓ_1-penalized linear regression on the binary outcome used as a feature selector; RFI, ranking by the gain-based feature importance of a random-forest gradient-boosting model; XGBRF, `XGBRFClassifier`; LinearSVC, linear support vector classifier with an ℓ_1 penalty, which acts as selector and classifier simultaneously; NuSVC, ν-support vector classifier with a radial basis function kernel. In the Component column, (S) marks the component acting as feature selector, (C) as classifier, and (S+C) a single object that performs both roles. Scale indicates how Optuna sampled the parameter: *log*, log-uniform over the interval; *int*, uniform over the integers of the interval; *float*, uniform over the interval.

Pipeline	Component	Hyperparameter	Range	Scale
LASSO–RF	`Lasso` (S)	`alpha`	$0.5/\sqrt{n}$ – $3/\sqrt{n}$	log
	`XGBRF` (C)	`n_estimators`	100 – 750	int
	`XGBRF` (C)	`max_depth`	1 – 10	int
LASSO–SVC	`LinearSVC` (S+C)	`C`	$0.5/\sqrt{n}$ – $3/\sqrt{n}$	log
RFI–RF	`XGBRF` (S)	`number_of_features`	1 – `X.shape[1]`	int
	`XGBRF` (S+C)	`n_estimators`	50 – 500	int
	`XGBRF` (S+C)	`max_depth`	1 – 10	int
RFI–SVC	`XGBRF` (S)	`number_of_features`	1 – `X.shape[1]`	int
	`XGBRF` (S)	`n_estimators`	50 – 500	int
	`XGBRF` (S)	`max_depth`	1 – 10	int
	`NuSVC` (C)	`nu`	0.05 – minority fraction	float

Notes to Table 2. n and `X.shape[1]` denote the number of rows and the number of columns of the training matrix passed to the search. Rows are patients for the global-based and distribution-based datasets and voxels for the voxel-based datasets, so the same expression yields different intervals across datasets; the number of columns reaches 20660 for the distribution-based datasets. The minority fraction is the proportion of the smaller outcome class among the training labels of the given fold. In the RFI–RF pipeline a single parameter dictionary configures both the model whose gain importances rank the features and the classifier fitted on the retained ones, which is why its rows are marked (S+C) rather than (C). All search spaces were explored with Optuna using the Gaussian-process sampler (`GPSampler`) with 20 trials per dataset-by-pipeline combination. The objective maximized was the *mean accuracy over the five inner cross-validation folds*, not AUC; the inner folds were generated by `StratifiedGroupKFold` grouped by `patient_id`. Settings that were fixed and not searched, one per line so that the long identifiers have somewhere to break:

```
Lasso(fit_intercept=True, max_iter=5000)
LinearSVC(penalty='l1', max_iter=5000, class_weight='balanced')
NuSVC(kernel='rbf', probability=True, class_weight='balanced')
XGBRFClassifier(random_state=42)
```

Class weighting was therefore applied in the two support vector branches only. The models were seeded with `random_state=42`, but `GPSampler()` was instantiated without a seed, so the sequence of 20 trials is not reproducible.

Table 3: Cross-validated performance of all 24 dataset-by-pipeline combinations in the MR CLEAN NO-IV cohort ($n = 141$), over five outer folds. Feature set: *global*, eight summary statistics of each radiomic feature over the thrombus, one row per patient; *distribution*, a ten-bin histogram of each radiomic feature over the thrombus, one row per patient; *voxel*, the radiomic feature values themselves, one row per voxel. ROI: *whole*, the dilated thrombus mask; *centerline*, the dilated skeleton of the thrombus mask. Selector (Sel.): LASSO, ℓ_1-penalized regression; L1, embedded selection by the non-zero coefficients of the classifier itself; RFI, gain-based feature importance of an `XGBRFClassifier`. Classifier (Class.): XGBRF, `XGBRFClassifier`; LinSVC, linear support vector classifier with an ℓ_1 penalty; NuSVC, ν-support vector classifier with a radial basis function kernel. In the body of the table the distribution-based feature set is abbreviated *distrib.* and the centerline ROI *cent.* AUC, area under the receiver operating characteristic curve; ACC, accuracy; IQR, interquartile range across the five folds; min–max, the smallest and largest value across the five folds; Med. n, median number of features retained by the selector; Null AUC, mean AUC of the permutation baseline; κ, the κ-type index of the observed AUC relative to that baseline; p, permutation p-value; q, the same p-value after Benjamini–Hochberg correction across all 24 combinations.

Feature set	ROI	Sel.	Class.	Median AUC [IQR]	AUC min–max	Median ACC	Med. n	Null AUC	κ	p	q (BH)
global	whole	LASSO	XGBRF	0.667 [0.631–0.690]	0.457–0.788	0.643	7	0.506	+0.284	0.0198	0.0528
global	whole	RFI	XGBRF	0.643 [0.641–0.667]	0.394–0.738	0.607	6660	0.505	+0.225	0.0198	0.0528
global	whole	L1	LinSVC	0.673 [0.656–0.697]	0.553–0.712	0.571	12	0.507	+0.307	0.0099	0.0396
global	whole	RFI	NuSVC	0.632 [0.594–0.662]	0.558–0.690	0.607	4552	0.501	+0.252	0.0198	0.0528
global	cent.	LASSO	XGBRF	0.406 [0.374–0.579]	0.354–0.616	0.500	10	0.491	−0.050	0.6832	0.9397
global	cent.	RFI	XGBRF	0.400 [0.383–0.418]	0.316–0.462	0.500	7582	0.499	−0.206	1.0000	1.0000
global	cent.	L1	LinSVC	0.415 [0.404–0.487]	0.379–0.581	0.464	53	0.506	−0.106	0.8614	0.9397
global	cent.	RFI	NuSVC	0.497 [0.399–0.549]	0.325–0.549	0.500	12845	0.502	−0.076	0.7921	0.9397
distrib.	whole	LASSO	XGBRF	0.367 [0.292–0.394]	0.256–0.453	0.500	1[a]	0.499	−0.291	1.0000	1.0000
distrib.	whole	RFI	XGBRF	0.651 [0.620–0.692]	0.567–0.694	0.571	8395	0.495	+0.297	0.0099	0.0396
distrib.	whole	L1	LinSVC	0.500 [0.500–0.574]	0.462–0.763	0.517	1	0.500	+0.120	0.0990	0.2160
distrib.	whole	RFI	NuSVC	0.636 [0.606–0.744]	0.579–0.810	0.643	18282	0.493	+0.359	0.0099	0.0396
distrib.	cent.	LASSO	XGBRF	0.471 [0.444–0.490]	0.433–0.500	0.536	1[a]	0.501	−0.068	0.8614	0.9397
distrib.	cent.	RFI	XGBRF	0.450 [0.438–0.510]	0.380–0.528	0.517	5068	0.510	−0.101	0.8119	0.9397
distrib.	cent.	L1	LinSVC	0.559 [0.387–0.577]	0.251–0.636	0.517	3	0.489	−0.014	0.5347	0.8871
distrib.	cent.	RFI	NuSVC	0.536 [0.428–0.561]	0.369–0.578	0.571	11871	0.502	−0.015	0.5545	0.8871
voxel[b]	whole	LASSO	XGBRF	0.551 [0.541–0.562]	0.512–0.581	0.554	9	0.499	+0.100	0.0099	0.0396
voxel[b]	whole	RFI	XGBRF	0.551 [0.541–0.560]	0.480–0.581	0.556	2066	0.499	+0.088	0.0099	0.0396
voxel[b]	whole	L1	LinSVC	0.531 [0.531–0.548]	0.502–0.596	0.545	63	0.500	+0.083	0.0495	0.1188
voxel[b]	whole	RFI	NuSVC	0.559 [0.552–0.562]	0.518–0.573	0.555	1543	0.502	+0.102	0.0099	0.0396
voxel[b]	cent.	LASSO	XGBRF	0.493 [0.477–0.503]	0.445–0.513	0.555	8	0.497	−0.022	0.7327	0.9397
voxel[b]	cent.	RFI	XGBRF	0.492 [0.486–0.506]	0.485–0.512	0.544	900	0.503	−0.013	0.6040	0.9059
voxel[b]	cent.	L1	LinSVC	0.514 [0.476–0.556]	0.435–0.583	0.506	92	0.497	+0.031	0.2673	0.4935
voxel[b]	cent.	RFI	NuSVC	0.515 [0.511–0.537]	0.475–0.574	0.536	199	0.502	+0.042	0.2277	0.4554

Notes to Table 3. Accuracy must be read against the majority-class accuracy of the development cohort, $82/141 = 0.582$; only 4 of the 24 combinations exceed it by median accuracy, and 86 of the 120 individual fold results (71.7%) fall below it. The class ratio itself varies across folds (0.448, 0.714, 0.536, 0.679, 0.536), so in the first fold the majority class is the absence rather than the presence of near-complete or complete reperfusion. The largest single fold result of the whole study is 0.810, obtained by distribution/whole/RFI/NuSVC in fold 4. Averaged over the eight whole-thrombus patient-level combinations, AUC per fold is 0.510, 0.661, 0.606, 0.587, 0.623, a spread within a single fold of up to 0.531; between-fold variability therefore exceeds between-method variability. The permutation null was obtained from 100 label permutations at the patient level per outer fold; with $m = 100$ permutations the smallest attainable p-value is $p = (0 + 1)/(100 + 1) = 0.0099$, which bounds the resolution of every permutation result in this table. The Null AUC, κ, p and q columns were computed from the *mean* AUC across folds, whereas the AUC column reports the *median*; the two summaries are not interchangeable and are shown side by side deliberately. 10 of the 24 combinations reach $p < 0.05$ uncorrected and 6 survive Benjamini–Hochberg correction for AUC, while none survives it for accuracy. No centerline combination survives correction; across the eight of them the mean AUC ranges from 0.396 to 0.495 against a null of 0.489–0.510, κ from -0.206 to -0.014 and p from 0.5347 to 1.0000.

[a] In these 2 combinations the ℓ_1 penalty shrank every coefficient to zero in every fold and the code substituted the index of the first column; the retained feature is a fallback, not one the selector identified as informative. Combinations with a median of one retained feature but no mark reached that median with genuinely non-zero coefficients.

[b] The unit of analysis of the voxel-based datasets is the voxel, not the patient, and their metrics are therefore *not* comparable with the remaining rows. Across the five folds the voxel/whole datasets contribute 396085 voxel-level predictions (91500, 79327, 75748, 82369, 67141) and the voxel/centerline datasets 8557 (1690, 1571, 1811, 1806, 1679), against 141 patient-level predictions (29, 28, 28, 28, 28) for every other dataset. The consequence is visible in the permutation null, whose standard deviation is 0.040–0.052 for the voxel-based combinations against 0.063–0.120 for all others, so that the voxel-based p-values are computed against a spuriously narrow null and are not comparable with the rest of the table. 3 voxel-based combinations are formally significant for AUC after correction; they must not be read as evidence for a signal.

Table 4: External validation in the MR CLEAN Registry cohort. Each pipeline was refitted on all 141 development patients with its tuned hyperparameters and applied once to the Registry cohort. CV, cross-validated median over the five outer folds of the same combination in Table 3; Reg., Registry; ACC, accuracy; ΔAUC, Registry AUC minus cross-validated median AUC, a negative value being a loss of discrimination on transfer; other abbreviations as in Table 3. The voxel-based datasets were not transferred externally and therefore have no row here, which is why the table has 16 rows rather than 24.

Feature set	ROI	Selector	Classifier	CV AUC	CV ACC	Reg. AUC	Reg. ACC	Δ AUC
global	whole	LASSO	XGBRF	0.667	0.643	0.492	0.490	−0.174
global	whole	RFI	XGBRF	0.643	0.607	0.471	0.490	−0.172
global	whole	ℓ_1 emb.	LinearSVC	0.673	0.571	0.483	0.494	−0.190
global	whole	RFI	NuSVC	0.632	0.607	0.481	0.470	−0.150
global	centerline	LASSO	XGBRF	0.406	0.500	0.483	0.535	+0.077
global	centerline	RFI	XGBRF	0.400	0.500	0.521	0.505	+0.121
global	centerline	ℓ_1 emb.	LinearSVC	0.415	0.464	0.506	0.521	+0.091
global	centerline	RFI	NuSVC	0.497	0.500	0.499	0.532	+0.002
distribution	whole	LASSO	XGBRF	0.367	0.500	0.494	0.441	+0.128
distribution	whole	RFI	XGBRF	0.651	0.571	0.482	0.448	−0.169
distribution	whole	ℓ_1 emb.	LinearSVC	0.500	0.517	0.504	0.472	+0.004
distribution	whole	RFI	NuSVC	0.636	0.643	0.482	0.460	−0.154
distribution	centerline	LASSO	XGBRF	0.471	0.536	0.514	0.441	+0.044
distribution	centerline	RFI	XGBRF	0.450	0.517	0.497	0.448	+0.047
distribution	centerline	ℓ_1 emb.	LinearSVC	0.559	0.517	0.499	0.550	−0.060
distribution	centerline	RFI	NuSVC	0.536	0.571	0.502	0.440	−0.034

Notes to Table 4. Across the 16 models the Registry AUC ranged from 0.471 to 0.521 with a median of 0.496, and 5 of them exceeded 0.50. Registry accuracy ranged from 0.440 to 0.550 with a median of 0.481, and no model reached the majority-class accuracy of the development cohort, 0.582. The sign of Δ follows the cross-validated value and nothing else: the 8 combinations whose cross-validated median AUC was above 0.500 all moved downwards on transfer, by −0.034 to −0.190, and the 8 that were at or below it all moved upwards, by 0.002 to 0.128. The upward movement is therefore a regression to chance of models that had performed below chance internally, not a gain in generalization, and it must not be presented as one. The only combinations for which the comparison is informative are the 6 that discriminated in cross-validation, that is reached a cross-validated median AUC of at least 0.600; all of them lose between 0.150 and 0.190 of AUC.

Table 5: Comparison with the reference pipeline of van Voorst et al. [1] The first row summarizes the present study; rows two to four were computed on the MR CLEAN NO-IV development cohort with the same five outer folds as in Table 3, so that the comparison is made within one dataset rather than across publications; the last row is the value reported in the original publication, on a different cohort and for a different endpoint. ICC, intraclass correlation coefficient: of the 107 features the reference pipeline extracts, the 52 with ICC ≥ 0.75 are retained and the rest discarded. $\mathrm{AUC}_{\mathrm{prob}}$ denotes the AUC computed from the predicted class probabilities, and is the quantity reported for the reference pipeline. Values are medians with the interquartile range in brackets, or means $\pm$ standard deviation across folds, as indicated in the note below. An em dash marks a quantity the row does not have: the pooled row spans combinations with different accuracies, and the published row reports neither a range nor an accuracy.

Model	Cohort	n	AUC	AUC min–max	Median ACC
Our best combinations, pooled	MR CLEAN NO-IV	141	0.656 [0.606–0.692]	0.457–0.810	—
Reference pipeline, recomputed	MR CLEAN NO-IV	141	0.481 ± 0.145	0.300–0.631	0.536
Our XGBRF on the 52 ICC-filtered features	MR CLEAN NO-IV	141	0.476	0.357–0.559	0.500
Our NuSVC on the 52 ICC-filtered features	MR CLEAN NO-IV	141	0.544	0.428–0.602	0.517
van Voorst et al., as published	MR CLEAN Registry	699	0.52	—	—

Notes to Table 5. The first row pools 25 fold results, 5 combinations $\times$ 5 outer folds, and its bracketed values are the interquartile range; the second row is a mean $\pm$ standard deviation over the 5 splits; rows three and four are medians over the 5 folds, with no further feature selection applied to the 52 ICC-filtered features. The last row is quoted from the publication: thrombus radiomics for eTICI $\geq$ 2B, a more permissive endpoint than the eTICI 2C–3 outcome of this article, as a mean over 100 bootstrap samples of 100 patients. The plain `AUC` column of the reference pipeline degenerates to exactly 0.50 in 4 of the five splits, because the fitted model assigns the majority class to every patient in those splits and its hard predictions therefore carry no ranking information. $\mathrm{AUC}_{\mathrm{prob}}$, computed from the predicted probabilities, is the only quantity of that pipeline that is comparable with the AUC values reported elsewhere in this article, and it is the one used here. By median accuracy the reference pipeline and both of our classifiers restricted to its 52 ICC-filtered features stay below the majority-class accuracy of 0.582, although the reference pipeline exceeds it in one individual split. Rows two to four were recomputed against the eTICI 2C–3 outcome of this article — the stored label column is named `TICI_2B-3`, but its per-fold positive counts are exactly the 82/141 of this study — and are directly comparable with the first row, whereas the published 0.52 of the last row was obtained for eTICI $\geq$ 2B in a different cohort and differs from them in endpoint and cohort as much as in pipeline.

Table 6: Ensemble analysis: the 15 configurations of discrete, that is binary, AdaBoost [2] applied to the stored prediction vectors of the 16 patient-level weak classifiers (four patient-level datasets × four pipelines). Each configuration was fitted on the five outer folds, giving 75 ensemble models. ACC ∅, mean accuracy of the configuration across the five folds; best weak classifier, the single pipeline in the configuration's pool with the highest mean accuracy, and its value; Δ, the mean accuracy of the ensemble minus that of the best weak classifier. A negative Δ means the ensemble is worse than the best model it combines. The configuration name states which of the four patient-level datasets form its pool, each contributing all four pipelines: `type1` pools all four datasets (16 weak classifiers), `type2` a whole-thrombus dataset with its own centerline counterpart (8), `type3` a single dataset (4), `type4` two datasets that are not such a pair (8) and `type5` all datasets but one (12). `distrib.` abbreviates the distribution-based dataset and *cent.* the centerline region of interest, as in Table 3. The two voxel-based datasets were excluded from every pool because their predictions are defined per voxel.

Configuration	ACC ∅	Best weak classifier	Its ACC ∅	Δ
`type1_all_featuresets`	0.5039	global/LASSO–XGBRF	0.6254	−0.121
`type2_distribution_pair`	0.5463	distrib./RFI–NuSVC	0.6246	−0.078
`type2_global_pair`	0.5611	global/LASSO–XGBRF	0.6254	−0.064
`type3_distribution`	0.5820	distrib./RFI–NuSVC	0.6246	−0.043
`type3_distribution_skeleton`	0.5323	distrib. cent./LASSO–XGBRF	0.5825	−0.050
`type3_global`	0.5825	global/LASSO–XGBRF	0.6254	−0.043
`type3_global_skeleton`	0.4887	global cent./LASSO–XGBRF	0.5108	−0.022
`type4_distribution__global`	0.5539	global/LASSO–XGBRF	0.6254	−0.071
`type4_distribution__global_skeleton`	0.5461	distrib./RFI–NuSVC	0.6246	−0.079
`type4_distribution_skeleton__global`	0.5468	global/LASSO–XGBRF	0.6254	−0.079
`type4_distribution_skeleton__global_skeleton`	0.5113	distrib. cent./LASSO–XGBRF	0.5825	−0.071
`type5_all_but_distribution`	0.5754	global/LASSO–XGBRF	0.6254	−0.050
`type5_all_but_distribution_skeleton`	0.5394	global/LASSO–XGBRF	0.6254	−0.086
`type5_all_but_global`	0.5318	distrib./RFI–NuSVC	0.6246	−0.093
`type5_all_but_global_skeleton`	0.5468	global/LASSO–XGBRF	0.6254	−0.079

Notes to Table 6. Δ is negative for all 15 configurations, between −0.022 and −0.121; no ensemble outperformed the best single pipeline in its own pool. The worst result belongs to `type1_all_featuresets`, the configuration that pools all feature sets. Across all 75 fitted ensemble models accuracy averages 0.543 ± 0.083 with a range of 0.414–0.714. The mechanism is documented by the fitted weights: in the all-feature-set configuration, 55 of the 80 boosting rounds (16 rounds × five folds) received a negative weight α, meaning that the weak classifier admitted in that round performed worse than chance on the reweighted data. The individual pipelines therefore do not make complementary errors, and the boosting assumption of weak classifiers reliably better than chance is not met. Labels were coded as $\{-1, +1\}$, the weight of a round was $\alpha = \frac{1}{2} \ln \frac{1-\varepsilon}{\varepsilon}$ with ε the error of the admitted classifier on the currently reweighted data, and each weak classifier could be admitted at most once, so a pool of k members yields at most k rounds. The weak classifiers were not retrained: the procedure is applied to models that had already been fitted, which makes it methodologically a form of stacking rather than boosting proper. The best weak classifier of `type3_global` and of `type1_all_featuresets` is the same pipeline, because the pool of the latter contains the pool of the former.

Table 7: Pairwise comparison of the pipelines by the corrected resampled t-test of Nadeau and Bengio [3]. The upper block compares the four pipelines of this study with one another on the whole-thrombus global-based dataset; the lower block compares them, and our two classifiers restricted to the reference pipeline's own 52 ICC-filtered features, against the reference pipeline of van Voorst et al. recomputed on the same cohort. Every model in the table was evaluated on the same 5 outer folds of the same 141 patients, which is what makes the paired test applicable. Because the training sets of k-fold cross-validation overlap, the per-fold differences are not independent and the ordinary paired t-test underestimates their variance; the correction multiplies the variance by $1/k + n_{\text{test}}/n_{\text{train}}$, here $1/5 + 28.2/112.8 = 0.4500$, which inflates the standard error 1.50-fold at df $= 4$. Fold sizes are taken from the data, not assumed. Mean ΔAUC is the mean of the 5 per-fold differences in AUC between pipeline A and pipeline B; a positive value favours A. t (uncorrected) is the ordinary paired statistic, t (corrected) the same quantity with the inflated variance. The two-sided critical value at $\alpha = 0.05$ and df $= 4$ is 2.776. Pipeline names combine selector and classifier as in Table 3.

Pipeline A	Pipeline B	Mean ΔAUC (A − B)	t (uncorrected)	t (corrected)	$\lvert t\rvert > t_{\text{crit}}$
LASSO–XGBRF	RFI–XGBRF	+0.030	+1.71	+1.14	no
LASSO–XGBRF	L1–LinSVC	−0.012	−0.42	−0.28	no
LASSO–XGBRF	RFI–NuSVC	+0.019	+0.39	+0.26	no
RFI–XGBRF	L1–LinSVC	−0.042	−1.28	−0.86	no
RFI–XGBRF	RFI–NuSVC	−0.010	−0.21	−0.14	no
L1–LinSVC	RFI–NuSVC	+0.031	+1.35	+0.90	no
LASSO–XGBRF	Reference pipeline	+0.165	+1.50	+1.00	no
RFI–XGBRF	Reference pipeline	+0.136	+1.23	+0.82	no
L1–LinSVC	Reference pipeline	+0.177	+2.15	+1.43	no
RFI–NuSVC	Reference pipeline	+0.146	+2.31	+1.54	no
XGBRF on ICC features	Reference pipeline	−0.010	−0.23	−0.15	no
NuSVC on ICC features	Reference pipeline	+0.046	+0.53	+0.35	no

Notes to Table 7. Not one of the 12 comparisons reaches significance: the largest corrected statistic in absolute value is 1.54, against a critical value of 2.776. The four pipelines of this study are therefore statistically indistinguishable from one another, and none of them is distinguishable from the reference pipeline either, in either direction. For the reference pipeline the compared quantity is AUC_{prob}, because its plain `AUC` column degenerates to exactly 0.50 in four of the five splits; see Table 5. The pooled row of Table 5 is deliberately absent here: those 5 combinations were picked by their results on these same folds, so testing them against the reference pipeline would be biased in favour of the present study. All four pipelines of the dataset are tested instead, whatever their outcome. With 5 folds the test has df $= 4$ and very little power, so these results are an absence of evidence for a difference, not evidence that the pipelines perform identically. The comparison that the data would support best, a DeLong test on the external ROC curves, was not possible because per-patient predicted scores on the Registry were not retained.

References

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